

Lane-Topology-Guided Motion Forecasting via Feasible Motion Primitive Selection

Sangjin Han[†], Hoseong Jung[†], Jeongtae Her[†], Changhyun Choi, H. Jin Kim^{*}

Abstract—Lane topology provides a strong structural prior for motion forecasting in autonomous driving systems designed to generalize across diverse road geometries. To be useful for safe downstream planning, predictions should be not only accurate but also feasible and diverse, covering a range of plausible future behaviors. However, existing predictors often neglect lane-topology consistency in lower-probability modes, producing trajectories that violate lane connectivity or traffic rules. In this paper, we propose a topology-aware forecasting framework designed to improve the structural reliability of multimodal predictions across varying road geometries. Specifically, we use a VQ-VAE to learn shape-aware motion primitives as discrete intention queries, capturing spatiotemporal patterns beyond endpoint-based intentions. We further introduce a feasibility-aware intention selector trained with a lane-topology prior to filter unreachable intention queries, guiding the decoder to prioritize topology-consistent intentions while preserving behavioral diversity. Extensive experiments on the Argoverse 2 dataset demonstrate that our method achieves prediction accuracy comparable to state-of-the-art baselines while outperforming them in feasibility and diversity metrics.

I. INTRODUCTION

Motion forecasting is a core component of autonomous driving systems as it predicts the future trajectories of surrounding agents based on their historical states and the map context [1], [2]. Due to inherent behavioral uncertainty and complex inter-agent interactions, future motions are intrinsically stochastic rather than deterministic [3]–[5]. Consequently, effective predictors must generate multiple plausible trajectories to capture diverse driving intentions [1]. Such multimodal predictions enable downstream planning modules to evaluate potential risks and ensure safe decision-making in dynamic traffic scenarios [6]–[8].

Existing approaches commonly rely on predefined intention priors, including trajectory anchors [9], [10] and target points [11], [12] to capture diverse driving intentions. While these priors facilitate multimodal coverage and stabilize training, they are typically designed independently of the specific scene context [9], [10]. As a result, their scene-agnostic design often fails to align with the underlying lane topology, producing structurally infeasible trajectories.

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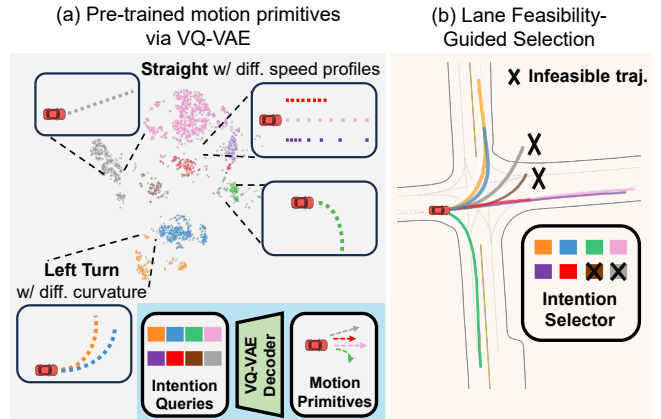


Fig. 1: **LAMP** improves the reliability of multimodal trajectory predictions. (a) A VQ-VAE provides a discrete set of intention queries; each query is decoded into a motion primitive representing a plausible future trajectory. (b) Infeasible intention queries are removed via lane-topology-guided selection, producing feasible and diverse set of trajectories for downstream planning.

Moreover, prevailing training objectives prioritize the displacement accuracy of the most probable mode [13]. Such objectives often overlook the feasibility of alternative modes. Therefore, lower-probability trajectories are more prone to violating lane constraints (e.g., off-road predictions), reducing their reliability for downstream planning. These limitations call for a topology-aware motion forecasting framework that enforces feasibility across all predicted modes while preserving multimodal diversity.

To address these challenges, we present **LAMP** (Lane-Aligned Motion Primitives), which improves the reliability of multimodal prediction sets by combining structured motion primitives with topology-aware intention selection. Rather than abstracting driving intentions as static endpoints, we model intention as a discrete set of learnable intention queries using a Vector Quantized Variational AutoEncoder (VQ-VAE) [14]. Each query decodes into a trajectory prototype, which we refer to as a motion primitive that captures recurring driving behavior patterns as illustrated in Fig. 1. This formulation thereby preserves the rich spatiotemporal dynamics often lost in endpoint-based methods. Beyond capturing dynamics, ensuring the alignment of motion primitives with the local road topology is essential to guarantee physical and logical feasibility. Accordingly, we introduce a feasibility-aware intention selector that filters out off-road or unreachable queries prior to decoding. This guides the model to generate a compact set of diverse trajectories that adhere to lane constraints for safety-critical planning.

We implement **LAMP** upon the Motion Transformer (MTR) architecture [1], within the UniTraj platform [15]. To evaluate both prediction accuracy and structural reliability, we report standard displacement metrics and further assess feasibility and diversity using established metrics and our refined variants. Experiments on the Argoverse 2 dataset [16] demonstrate that **LAMP** achieves performance comparable with strong baselines on standard displacement metrics while substantially improving feasibility and diversity.

In summary, our contributions are threefold:

- We propose **LAMP**, a topology-aware motion forecasting framework that improves the reliability of multimodal prediction sets via structured motion primitives and feasibility-aware intention selection.
- We formulate driving intention as discrete intention queries via a VQ-VAE, where each query decodes to a trajectory prototype serving as a motion primitive that captures rich spatiotemporal dynamics.
- We introduce a lane-topology-guided intention selector and show improved feasibility and diversity while maintaining competitive displacement accuracy.

II. RELATED WORK AND BACKGROUND

A. Modeling Multi-Modality in Motion Forecasting

Generating diverse and plausible future trajectories remains a central challenge in motion forecasting due to the inherent stochasticity of drivers. Early works sample multiple futures from a latent generative models [3]–[5], [17], but may suffer from mode collapse or structurally inconsistent predictions. Anchor-based approaches introduce predefined intention priors (e.g., trajectory templates or target candidates) to improve multimodal coverage [10]–[12], [18]. These static anchors enhance training stability and encourage diverse predictions, but their scene-agnostic design may limit alignment with complex lane topology. More recently, query-based methods [1], [19]–[21] adopt learnable queries to dynamically interact with scene context. For instance, MTR [1] constructs intention queries derived from a fixed set of intention points to aggregate local map context for multimodal prediction. This creates a trade-off: fixed queries can limit adaptability, while fully dynamic querying can be harder to optimize. Our method addresses this trade-off by leveraging primitive-based intention queries and a lane-topology-guided intention selector to produce feasible yet diverse multimodal predictions.

B. Intention Modeling for Reliable MultiModal Prediction

1) *Intention Representation*: Modeling driving intention requires an abstraction that captures meaningful motion semantics. Endpoint-based methods [11], [12] represent intention through discrete target candidates, treating the final destination as the primary behavioral indicator. While effective for goal-directed reasoning, they overlook trajectory shape and temporal evolution, limiting their ability to distinguish nuanced behaviors.

An alternative is to represent intention with motion primitives instead of endpoints. Vector quantization models, such

as VQ-VAE [14], introduce a discrete bottleneck by learning a codebook whose codes can be decoded into trajectory prototypes. This discrete representation has been explored in trajectory forecasting to model multimodality with a finite set of latent codes [22], [23]. However, learning a well-utilized codebook can be challenging due to optimization issues such as gradient collapse and codebook collapse. Recent variants address these issues and improve training stability, including NSVQ [24] and FSQ [25]. In our setting, the codebook is learned from a large corpus of collected driving trajectories, so we adopt NSVQ for stable codebook learning.

2) *Feasibility-Aware Selection*: Beyond representation, ensuring the feasibility of the entire prediction set is critical for downstream planning [26]. In practice, low-probability modes are more likely to violate map constraints, reducing the reliability of multimodal outputs [2]. Prior work commonly enforces feasibility via trajectory-level auxiliary objectives [6] or post-hoc heuristic corrections [11]. However, they do not explicitly model feasibility as a selection criterion over multimodal hypotheses. In contrast, we learn a lane-topology-guided intention selector that prunes unreachable or off-road intention queries.

III. METHODS

We propose **LAMP** (Lane-Aligned Motion Primitives), a topology-aware motion forecasting framework that improves the reliability of multimodal prediction sets using structured motion primitives and feasibility-aware intention selection. The overall framework is illustrated in Fig. 3. The remainder of this section details each component. Sec. III-A describes the task formulation and the backbone architecture. Sec. III-B describes the learning of the discrete intention queries. Sec. III-C presents the lane-topology-guided intention selector for feasibility-aware selection. Finally, Sec. III-D outlines the training and inference pipeline.

A. Problem Formulation and Backbone Architecture

1) *Task Definition*: The goal of motion forecasting is to predict the future trajectory of a target agent over a horizon T_f , given historical context. We adopt vectorized representations [2], where the input consists of agent histories $A \in \mathbb{R}^{N_a \times T_h \times C_a}$ and map polylines $M \in \mathbb{R}^{N_m \times N_p \times C_m}$. Here, N_a , T_h , and C_a denote the number of agents, past timesteps, and motion features, while N_m , N_p , and C_m denote the number of polylines, points per polyline, and point attributes, respectively. Following the agent-centric setting [18], inputs are normalized to the target’s coordinate frame. The model generates L trajectories $\mathcal{Y} = \{Y_l, p_l\}_{l=1}^L$, where $Y_l \in \mathbb{R}^{T_f \times 2}$ is the predicted path and p_l denotes the confidence.

2) *Scene Context Encoding*: We utilize the Motion Transformer (MTR) [1] as the scene encoder. Agent histories A and map polylines M are first embedded via PointNet-style MLPs to obtain initial features $F_A^{(0)}$ and $F_M^{(0)}$ [27]:

$$F_A^{(0)} = \phi(\text{MLP}(A)), \quad F_M^{(0)} = \phi(\text{MLP}(M)), \quad (1)$$

where ϕ denotes max-pooling. To model local lane topology, MTR employs a local Transformer encoder that restricts

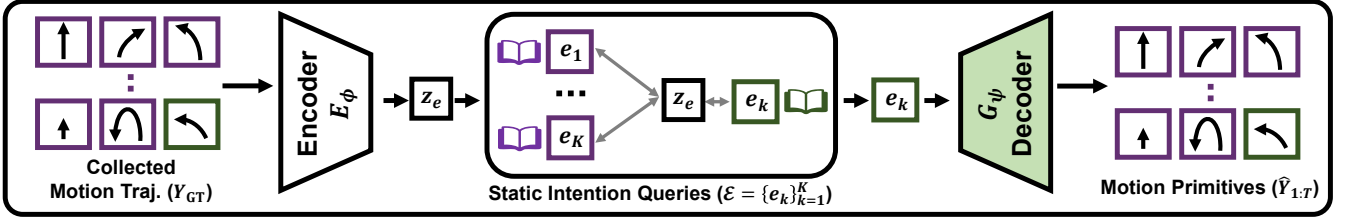


Fig. 2: Learning motion primitives with VQ-VAE [14]. Collected motion trajectories Y_{GT} are encoded into a discrete codebook of intention queries to represent diverse driving intentions with finite codes. Each code is decoded to a trajectory prototype $\hat{Y}_{1:T}$, which serves as a motion primitive in **LAMP**.

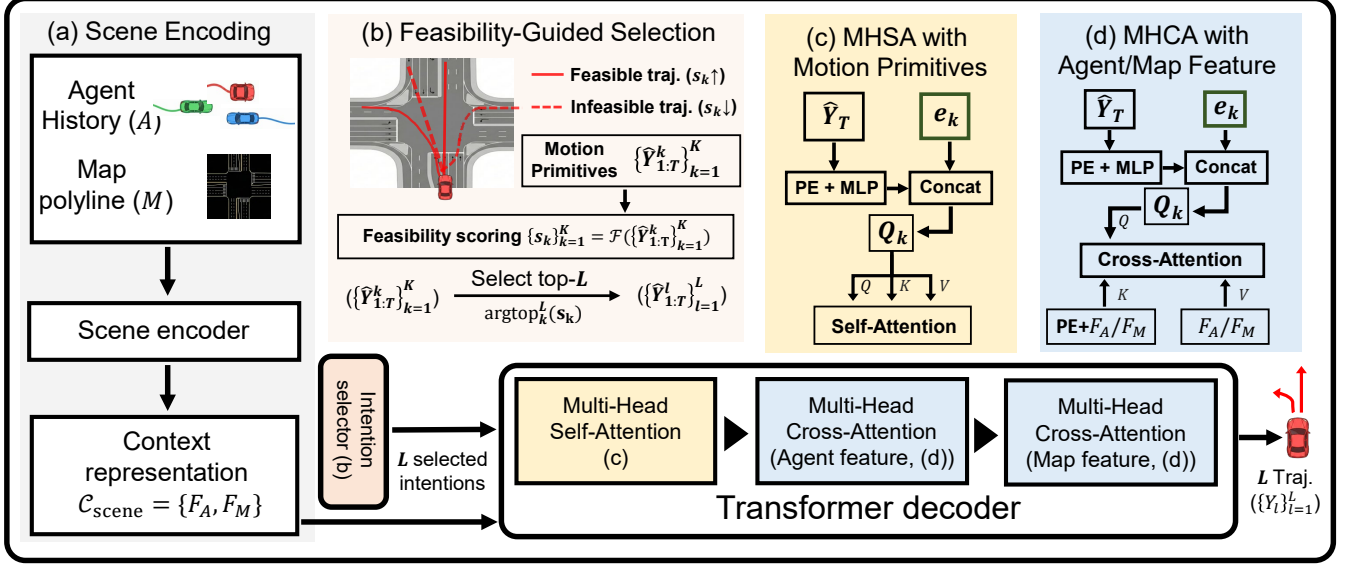


Fig. 3: Overview of the **LAMP** framework. (a) Scene context is encoded from agent histories and map polylines. (b) A feasibility-guided intention selector evaluates motion primitives and retains L topology-consistent candidates from K hypotheses. The selected intention embeddings and scene features are fed into a Transformer decoder. (c) Multi-head self-attention (MHSA) refines motion queries formed from static intention queries and motion primitives, followed by (d) cross-attention (MHCA) with agent and map context to generate L trajectories.

attention to the k -nearest neighbors $\Omega(i)$:

$$F^{(l+1)}[i] = \text{MHSA}(h_i, \{h_j\}_{j \in \Omega(i)}), \quad (2)$$

where h_i denotes the query token feature at the l -th layer, and $\{h_j\}$ represents the key and value features from the local neighborhood. An auxiliary dense motion prediction (DMP) loss further enhances agent features:

$$S_{\text{future}} = \text{MLP}(F_A), \quad \mathcal{L}_{\text{DMP}} = |S_{\text{future}} - S_{\text{GT}}|. \quad (3)$$

This produces the final context representation $\mathcal{C}_{\text{scene}} = \{F_A, F_M\}$ used in our subsequent decoding modules.

3) *Decoding Trajectories*: The standard MTR decoder employs a set of learnable queries to interact with the scene context via cross-attention. Specifically, it defines K static intention points $\mathcal{I} = \{I_k\}_{k=1}^K$ (e.g., via k -means clustering). The intention query Q_k is generated by encoding these points with positional embedding (PE):

$$Q_k = \text{MLP}(\text{PE}(I_k)). \quad (4)$$

These queries are then fed into the Transformer decoder to aggregate context features. The final output is modeled as a Gaussian Mixture Model (GMM). For a given time step t ,

the probability distribution of the agent's position o is:

$$P_t(o) = \sum_{k=1}^K p_k \cdot \mathcal{N}(o | \mu_k^t, \Sigma_k^t). \quad (5)$$

The model is trained end-to-end by minimizing the negative log-likelihood loss for the best-matching mode (Top-1) combined with the auxiliary regression loss:

$$\mathcal{L}_{\text{BASE}} = -\log P(\hat{Y}_{\text{GT}}) + \lambda_{\text{DMP}} \mathcal{L}_{\text{DMP}}. \quad (6)$$

In the following sections, we introduce our intention queries and feasibility-aware selection on top of this backbone.

B. Learning Intention Queries and Motion Primitives

We learn a discrete codebook of intention queries using a NSVQ-based VQ-VAE [14], [24], replacing the endpoint-based intention priors derived via K -means clustering in MTR [1]. Fig. 2 illustrates the quantization pipeline. Given a collected motion trajectory $Y_{\text{GT}} \in \mathbb{R}^{T \times 2}$, an encoder E_ϕ maps Y_{GT} into a latent representation z_e . We maintain a learnable codebook $\mathcal{E} = \{e_k\}_{k=1}^K \subset \mathbb{R}^{D_e}$, where K is the codebook size and D_e the embedding dimension. The latent z_e is quantized to the nearest code vector z_q :

$$z_q = e_k \quad \text{where } k = \underset{j}{\operatorname{argmin}} \|z_e - e_j\|_2^2. \quad (7)$$

TABLE I: Argoverse 2 [16] validation results for main comparisons (Sec. IV-B). We report accuracy (b-minADE₆, b-minFDE₆, minADE₆, minFDE₆), feasibility (DAC, FR), and diversity (APD₆, FPD₆, DwF). Bold entries denote the best result for each metric within each block.

	Models	Accuracy				Feasibility		Diversity		
		b-minADE ₆ ↓	b-minFDE ₆ ↓	minADE ₆ ↓	minFDE ₆ ↓	DAC↑	FR↑	APD ₆ ↑	FPD ₆ ↑	DwF↑
Main Results	LAMP (Ours)	1.345	2.281	0.904	1.785	0.942	0.774	6.401	16.232	12.559
	MTR [1]	1.315	2.127	0.842	1.667	0.925	0.665	4.109	10.963	7.811
	Wayformer [21]	1.420	2.281	0.804	1.664	0.905	0.691	4.157	11.151	8.173
	Forecast-MAE [20]	1.379	2.125	0.751	1.492	0.937	0.692	2.983	7.728	5.657
	EMP [28]	1.385	2.141	0.749	1.503	0.934	0.684	2.936	7.608	5.524
	Autobot [17]	1.512	2.389	0.846	1.720	0.937	0.699	3.567	9.163	6.829

A decoder G_ψ reconstructs the trajectory prototype $\hat{Y}_{1:T}$ from z_q . We adopt NSVQ [24] to alleviate reliance on auxiliary commitment losses and to address the gradient-collapse issue in vector quantization. Accordingly, we optimize the quantization module with the reconstruction objective:

$$\mathcal{L}_{VQ} = \|Y_{GT} - \hat{Y}_{1:T}\|_2^2, \quad (8)$$

where NSVQ provides nonzero gradients for both encoder outputs and codebook vectors during training.

This results in a discrete codebook $\mathcal{E}$ whose codes decode to trajectory prototypes (motion primitives) capturing diverse and realistic driving patterns. For each code e_k , we form an intention query Q_k by combining the code embedding with a spatial grounding term. Specifically, we extract the endpoint $\hat{Y}_T$ from its decoded trajectory prototype (via the frozen decoder G_ψ) and concatenate its positional encoding with the code embedding:

$$Q_k = \text{Concat}(e_k, \text{MLP}(\text{PE}(\hat{Y}_T))). \quad (9)$$

The enriched queries $\mathcal{Q} = \{Q_k\}_{k=1}^K$ are then fed into the motion decoder to interact with the scene context.

C. Feasibility-Aware Intention Selection

Given the K motion primitives $\{\hat{Y}_{1:T}^k\}_{k=1}^K$ decoded from intention queries, we learn an intention selector that ranks and retains feasible modes before final decoding. Passing all K hypotheses to the Transformer decoder can mix infeasible intentions into the attention context, reducing the reliability of the final prediction set. We therefore select only the top- L feasible intentions and forward them to the decoder. The selector is trained with a lane-topology prior $P_{\text{lane}}(k)$ and predicts a score distribution $P_{\text{pred}}(k)$ given the scene context, where we use $s_k = P_{\text{pred}}(k)$ for top- L selection.

We define a feasibility scoring function $\mathcal{F}(\cdot)$ (Fig. 3(b)) to provide a topology-aware supervision signal. In this work, $\mathcal{F}(\cdot)$ is instantiated using a lane-topology prior. Specifically, we automatically label a set of reachable lanes from the current agent pose by traversing HD-map lane connectivity. We compute a distance d_k between each hypothesis $\hat{Y}_T^k$ and the reachable-lane set and form a soft teacher distribution:

$$P_{\text{lane}}(k) = \frac{\exp(-\alpha d_k)}{\sum_{j=1}^K \exp(-\alpha d_j)}, \quad (10)$$

where α controls the sharpness of the distribution. The selector is trained to match the lane prior via:

$$\mathcal{L}_{\text{IS}} = D_{\text{KL}}(P_{\text{lane}} \| P_{\text{pred}}). \quad (11)$$

The selected intention embeddings are combined with scene features to form motion queries. These queries are refined by a Transformer decoder through multi-head self-attention and cross-attention (Fig. 3(c,d)).

D. Training and Inference Pipeline

1) *Multi-Stage Training Strategy*: Training proceeds in two stages. In Stage 1, we train the VQ-VAE via $\mathcal{L}_{VQ}$ to learn a discrete codebook of intention queries and decoded motion primitives. We freeze the VQ-VAE codebook and decoder and use the resulting intention queries as fixed prototypes. In Stage 2, we jointly train the forecasting backbone and intention selector end-to-end by minimizing:

$$\mathcal{L}_{\text{TOTAL}} = \mathcal{L}_{\text{BASE}} + \lambda_{\text{IS}} \mathcal{L}_{\text{IS}}. \quad (12)$$

2) *Inference*: Our model generates L trajectory hypotheses with confidence scores $\{p_k\}_{k=1}^L$. For evaluation, we select $L_{\text{eval}} = 6$ final predictions using non-maximum suppression (NMS). Following MTR [1], we greedily pick the highest-confidence hypothesis and suppress other candidates whose endpoint distance to a selected one is below 2.5m, until L_{eval} trajectories are obtained. All metrics are computed on this NMS-selected set unless otherwise specified.

IV. EXPERIMENTS

A. Experimental Setup

We evaluate our framework on the Argoverse 2 Motion Forecasting Dataset [16], which contains 250K scenarios with diverse road geometries and agent interactions. The task is to predict the 6-second trajectories (T_f) of a target agent given 5 seconds of history (T_h). Our implementation is built upon UniTraj [15] framework with the official MTR [1] backbone. The model is trained on 4 NVIDIA RTX A5000 GPUs. For learning intention queries, we train NSVQ-based VQ-VAE with latent dimension $D_e = 512$ and codebook size $K = 64$ using Adam optimizer with a learning rate of 10^{-4} for 100 epochs (batch size 256). We then train the main forecasting model using AdamW optimizer with a learning rate of 10^{-4} for 32 epochs (batch size 64). We set the number of selected intentions to $L = 16$, the lane-prior temperature to $\alpha = 1.0$, and the selector loss weight to $\lambda_{\text{IS}} = 50$.

We compare **LAMP** against strong baselines in the motion forecasting literature. Specifically, we include **MTR** [1], **Wayformer** [21], **Forecast-MAE** [20], **EMP** [28], and **Autobot** [17]. All results are reported on the validation set following the standard evaluation protocol.

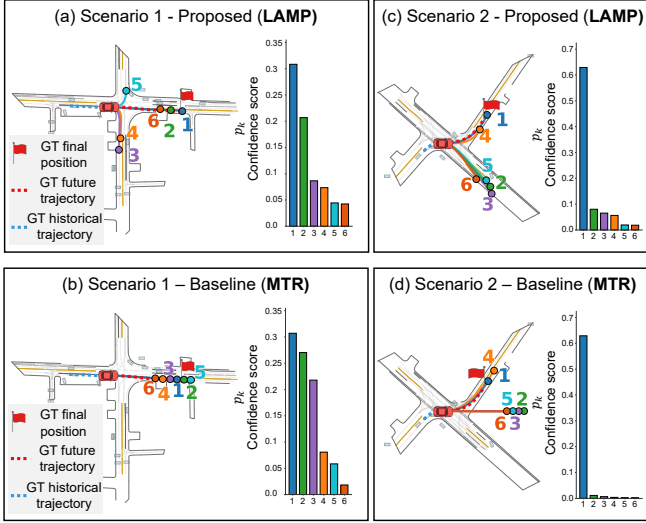


Fig. 4: Qualitative comparison of **LAMP** and **MTR** on two intersection scenarios. Each panel visualizes multimodal predictions with confidence scores p_k , along with agent history, ground-truth future, and final position. Scenario 1 highlights improved diversity of **LAMP**, while Scenario 2 shows improved feasibility by reducing off-road predictions compared to **MTR**.

We report standard displacement metrics for prediction accuracy (**b-minADE₆**, **b-minFDE₆**, **minADE₆** and **minFDE₆**) [16], [29]. For structural reliability, we report two feasibility metrics with different strictness: **DAC** [30] measures whether trajectories stay within the drivable area, while **FR** further checks traffic-rule constraints (e.g., lane connectivity). For diversity, we include self-distance metrics used in prior work [31], [32] (**APD₆**, **FPD₆**), and introduce **DwF** (Diversity-while-Feasible), a feasibility-aware diversity metric. DwF measures diversity by averaging pairwise endpoint distances over the feasible subset.

B. Main Results

Table I presents the quantitative results on Argoverse 2. **LAMP** substantially improves the feasibility and diversity of multimodal prediction sets while maintaining competitive prediction accuracy. Compared to strong baselines, our gains are most pronounced in feasibility under stricter constraints. In particular, DAC primarily captures whether a trajectory remains within the drivable area, whereas FR further evaluates lane topology and traffic-rule consistency. Since downstream planners typically reject hypotheses that are unreachable or rule-violating even when they are nominally on-road, improvements in FR are more indicative of planner-relevant feasibility. Accordingly, **LAMP** exhibits a larger improvement in FR than in DAC, indicating that **LAMP** reduces not only off-road trajectories but also traffic rule-violating motions. **LAMP** also improves diversity in a planner-relevant manner. For downstream planning, diversity is useful only when the multimodal set remains feasible. Thanks to the learned motion primitives from NSVQ-based VQ-VAE, **LAMP** increases overall diversity (APD₆/FPD₆), indicating improved intention coverage. This combination provides a reliable multimodal prediction set that offers diverse yet feasible hypotheses for downstream planners.

TABLE II: Comparison of **LAMP** and **LAMP+LoRA** on the Argoverse 2 validation set. We report feasibility and diversity metrics.

Model	DAC↑	FR↑	APD ₆ ↑	FPD ₆ ↑	DwF↑
LAMP	0.942	0.774	6.401	16.232	12.559
LAMP + LoRA	0.983	0.794	3.943	10.402	8.092

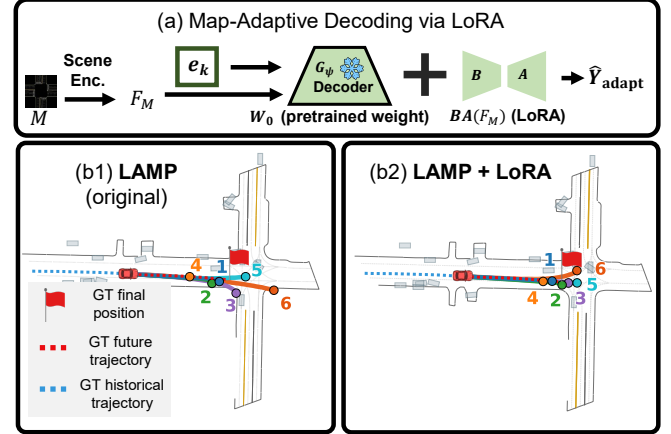


Fig. 5: LoRA-based map conditioning for motion-primitive reconstruction. (a) We apply LoRA [33] to the NSVQ decoder to incorporate map features. (b) **LAMP+LoRA** increases feasibility but yields less diverse multimodal predictions than **LAMP**.

C. Qualitative Analysis

Fig. 1 visualizes the learned intention space and feasibility-aware selection in **LAMP**. The t-SNE projection (Fig. 1a) shows that decoded intention queries exhibit interpretable driving patterns with varying curvatures and speeds. This suggests that the codebook effectively captures diverse trajectory shapes beyond simple endpoint-based priors. Subsequently, the intention selector filters out structurally infeasible primitives (Fig. 1b), enhancing the reliability of the prediction set. Fig. 4 further demonstrates **LAMP**'s qualitative advantages over **MTR** on intersection scenarios. While **MTR** often concentrates on a single dominant mode (Scenario 1) or assigns confidence to off-road trajectories (Scenario 2), **LAMP** generates diverse, plausible maneuvers that strictly adhere to the lane topology. These results validate that our approach improves structural reliability while preserving multimodal coverage.

D. Trade-off Analysis: Map-Adaptive Decoding via LoRA

While **LAMP** learns scene-agnostic intention queries and decodes them into motion primitives, we additionally explored injecting map priors into the NSVQ decoder via Low-Rank Adaptation (LoRA) [33]. As illustrated in Fig. 5(a), we freeze the pre-trained decoder weights W_0 and inject a lightweight low-rank update modulated by map features F_M :

$$h = W_0 x + \Delta W(F_M) x = W_0 x + BA(F_M) x, \quad (13)$$

which yields map-adaptive motion primitives $\hat{Y}_{\text{adapt}}^k$ during reconstruction. Table II shows that map-adaptive variant (**LAMP+LoRA**) improves feasibility metrics, while it reduces diversity metrics. Fig. 5(b) further confirms this trend:

compared to **LAMP**, **LAMP+LoRA** produces more lane-consistent predictions but a more concentrated set of hypotheses. As shown in Fig. 5, early map conditioning induces mode contraction; intention queries become heavily biased toward dominant lane-following behaviors, collapsing multimodal hypotheses. Due to this feasibility–diversity trade-off, we exclude LoRA from the final **LAMP** model, leaving diversity-preserving scene adaptation as future work.

V. CONCLUSION

In this paper, we presented **LAMP**, a topology-aware motion forecasting framework that improves the reliability of multimodal predictions. By leveraging an NSVQ-based VQ-VAE, **LAMP** encodes driving behaviors as reusable motion primitives that capture trajectory patterns beyond endpoint-based intentions. Crucially, our feasibility-aware selector evaluates intention queries against lane topology prior to decoding, filtering infeasible modes without sacrificing multimodal diversity. Experiments on Argoverse 2 demonstrate that **LAMP** substantially improves feasibility and diversity while maintaining competitive displacement accuracy. Future work will focus on closed-loop downstream planning and real-vehicle validation.

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